

WILLMAR, MN, USA

WMO: 720386

Lat: 45.117N

Lon: 95.133W

Elev: 344

StdP: 97.26

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 00125

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-23.6	-30.1	0.2	-25.8	-27.2	0.3	-23.4	14.4	-10.4	12.7	-9.8	2.7	300	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.1	22.6	30.1	22.1	28.7	21.4	24.9	29.7	23.8	28.2	22.7	27.1	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.5	27.7	22.4	17.8	27.1	21.3	16.6	25.8	77.4	29.9	72.9	27.8	68.8	27.2	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.6	9.0	DB	-29.7	34.9	3.6	1.2	-32.3	35.8	-34.4	36.5	-36.5	37.1	-39.1	38.0	(4)
(5)			WB	-29.9	26.8	3.5	1.3	-32.4	27.7	-34.5	28.5	-36.5	29.2	-39.0	30.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-10.8	-8.2	-1.0	7.2	14.1	20.0	22.1	20.6	16.7	8.9	0.4	-7.2	(6)	
	DBStd	12.79	7.62	7.35	7.15	5.55	4.89	3.73	3.08	3.02	4.70	5.23	6.30	6.52	(7)	
	HDD10.0	2545	644	511	344	117	17	0	0	0	4	83	292	533	(8)	
	HDD18.3	4481	903	744	598	336	148	25	6	13	86	294	538	792	(9)	
	CDD10.0	1443	0	0	4	32	145	299	374	327	206	50	4	0	(10)	
	CDD18.3	336	0	0	0	1	18	74	122	82	37	3	0	0	(11)	
	CDH23.3	2981	0	0	1	23	223	662	1077	630	323	40	1	0	(12)	
	CDH26.7	869	0	0	0	6	76	206	328	153	91	9	0	0	(13)	
(14) Wind	WSAvg	4.0	4.3	4.2	4.4	4.8	4.5	3.8	3.0	2.8	3.4	4.1	4.2	4.1	(14)	
(15) Precipitation	PrecAvg	754	20	20	40	73	82	125	96	99	77	64	38	20	(15)	
	PrecMax	974	48	47	85	175	178	218	169	190	213	174	110	52	(16)	
	PrecMin	545	2	7	12	20	26	64	21	12	15	16	1	5	(17)	
	PrecStd	110	14	12	20	38	37	48	37	46	48	40	33	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	11.2	19.9	26.3	32.4	33.5	33.1	32.3	32.2	27.7	19.0	8.7	(19)	
		MCWB	3.1	9.0	14.0	16.1	18.5	22.4	24.8	23.8	21.6	17.4	13.6	5.7	(20)	
	2%	DB	2.6	6.3	15.8	22.5	28.7	31.2	32.0	29.9	29.0	23.0	15.9	5.1	(21)	
		MCWB	1.2	4.2	11.9	13.3	17.7	21.3	24.4	23.4	21.0	15.7	10.5	2.9	(22)	
	5%	DB	1.2	3.0	12.0	19.0	26.0	28.8	29.9	27.9	27.3	20.1	12.2	2.7	(23)	
		MCWB	0.3	1.7	9.4	11.7	16.4	20.4	23.2	21.8	19.8	13.9	8.5	1.4	(24)	
	10%	DB	-0.9	2.2	7.9	16.4	22.7	27.3	28.0	27.2	24.8	17.4	8.8	1.4	(25)	
		MCWB	-1.3	1.1	5.9	10.2	15.1	19.3	21.9	21.1	18.6	12.5	6.2	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	8.8	15.6	17.3	21.8	24.9	27.0	25.7	24.2	19.7	14.7	6.6	(27)	
		MCDB	5.9	11.2	17.7	23.6	28.5	30.7	30.9	29.9	28.7	24.1	18.0	8.5	(28)	
	2%	WB	1.6	4.1	12.0	14.6	19.4	23.4	25.4	24.2	22.6	17.1	11.1	3.2	(29)	
		MCDB	2.6	5.7	14.6	20.3	25.4	28.7	30.2	28.2	27.4	21.3	14.0	4.8	(30)	
	5%	WB	0.3	2.4	8.7	12.6	17.8	21.9	24.1	23.1	21.3	15.0	8.4	1.7	(31)	
		MCDB	1.1	3.5	11.0	17.7	23.1	26.6	28.4	27.0	25.6	19.3	11.5	2.7	(32)	
	10%	WB	-1.1	0.8	5.8	11.0	16.2	20.4	23.1	21.9	19.7	12.8	6.1	0.4	(33)	
		MCDB	-0.7	1.9	8.3	16.1	21.1	25.4	27.4	25.8	23.7	16.9	8.7	1.4	(34)	
(35) Mean Daily Temperature Range	MDBR		9.0	9.6	9.1	10.9	11.4	10.6	10.6	10.7	11.8	10.8	9.0	8.3	(35)	
	5% DB	MCDBR	9.9	9.8	13.7	16.3	16.0	13.2	11.7	11.9	13.9	15.5	13.7	10.1	(36)	
		MCWBR	8.6	8.2	9.5	9.2	7.7	6.2	6.2	6.6	7.2	8.6	9.2	8.4	(37)	
	5% WB	MCDBR	9.2	9.2	13.1	13.7	12.7	11.0	10.6	10.7	11.8	12.9	12.7	9.4	(38)	
		MCWBR	8.1	7.6	9.4	8.7	7.4	6.4	6.3	6.7	7.1	8.1	9.0	8.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.260	0.275	0.307	0.363	0.388	0.392	0.392	0.402	0.363	0.315	0.279	0.255	(40)	
	taud		2.449	2.404	2.428	2.327	2.321	2.363	2.354	2.312	2.400	2.528	2.538	2.476	(41)	
	Ebn at Noon		873	919	930	896	881	876	871	848	856	863	845	844	(42)	
	Edh at Noon		69	87	98	119	124	120	119	120	101	77	63	60	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.77	2.84	3.79	4.70	5.41	6.02	6.48	5.37	4.15	2.66	1.72	1.26	(44)	
	RadStd		0.21	0.33	0.36	0.43	0.41	0.41	0.23	0.44	0.31	0.37	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page